Data structures and algorithms are interconnected, they need each other.

- Data structures are the substrate, a way to organize an area of memory to represent data.
- Algorithms are procedures, sequence of instructions aimed to transform data.

Without algorithms to transform them, data structures would just be bits stored on a memory chip; without data structures to operate on, most algorithms wouldn't even exist.

Every data structure, moreover, implicitly defines algorithms that can be performed on it: for instance, methods to add, retrieve and remove elements to/from the data structure.

Some data structure is actually defined exactly with the purpose of enabling one or more algorithms to run efficiently on them: think of hash tables and search by key[1].

So, when we use algorithm and data structure as synonyms, it's just because in that particular context one implies the other: for instance, when we describe a *DS*, for that description to be meaningful and accurate, we necessarily need to describe its methods (i.e. algorithms).

1.2 Data Structures

To start with our journey, we first need to agree on a common language to describe and evaluate algorithms.